Marginal zone lymphoma

Antigen-driven lymphoma — extranodal (MALT), splenic and nodal MZL share a CD5–/CD10– phenotype but not a treatment. **Treat the driver first** (H. pylori, HCV, C. psittaci), irradiate localized disease with curative intent, and reserve systemic therapy for disseminated or symptomatic disease

Compiled from primary trial reports and consensus guidelines (NCCN; WHO-HAEM5; EGILS) verified against PubMed, September 2026.

ENTITIES — WHO-HAEM5

Extranodal MZL of MALT *most common*	Arises in mucosa made lymphoid by **chronic antigen**. **Stomach most common**; then ocular adnexa, lung, salivary gland, thyroid, skin, breast, dura, small bowel. Lymphoepithelial lesions. Usually localized — **often curable locally**
Splenic MZL	Spleen + marrow (intrasinusoidal) ± blood; **splenomegaly, cytopenias, villous lymphocytes**, small IgM paraprotein, autoimmune cytopenias. Nodes uncommon. **HCV**-linked subset. del(7q), **NOTCH2, KLF2** mutations
Nodal MZL	Nodes only (no extranodal or splenic primary). Behaves and is **treated like FL**
Primary cutaneous MZL	**Split off as its own entity in WHO-HAEM5**; very indolent, skin-directed therapy. **Borrelia** link in European series

All are CD5– CD10– small B cells, but genetics differ by site: **t(11;18)(q21;q21) BIRC3::MALT1** in gastric and lung MALT; **TNFAIP3** loss in ocular adnexa; **GPR34** in salivary gland; **FAS** in cutaneous MZL; **KMT2D, NOTCH2, KLF2** in nodal MZL. Also t(14;18)(q32;q21) **IGH::MALT1**, t(1;14) IGH::BCL10, t(3;14) IGH::FOXP1.

WORKUP

- **Every MZL: HCV antibody + RNA, HBsAg + anti-HBc, HIV**, SPEP/immunofixation (paraprotein common), LDH, β2M
- **Gastric:** EGD with multiple biopsies (**map the stomach**), ***H. pylori*** by histology **+** stool antigen or breath test (serology if on PPI), **EUS** for depth and perigastric nodes, **FISH for t(11;18)**
- **Splenic:** smear (villous lymphocytes), flow, marrow biopsy, CT; DAT and autoimmune screen
- **PET/CT** is less reliable than in FL (MALT and SMZL are variably FDG-avid) — CT is acceptable; **a hot focus raises concern for transformation**
- Autoimmune history (Sjögren, Hashimoto) and site-specific exams: ophthalmology, ENT, thyroid ultrasound

ANTIGEN DRIVER → ANTIMICROBIAL FIRST

Site	Driver	Antimicrobial therapy & evidence
Stomach	***H. pylori***	**Eradication is first therapy** for localized disease. Remission **77.5%** (32 studies, 1,408 pts): stage I 78% vs II1 56%; mucosa/submucosa 82% vs deeper 54%; **t(11;18)+ only 22% vs 78%**
Ocular adnexa	***Chlamydia psittaci*** (regional)	**Doxycycline** 3 wk: ORR 65%, 5-yr PFS 55% (phase 2, stage I); better if Cp eradicated (86% vs 47%). Cp DNA in 89% of that Italian cohort — prevalence varies widely by region
Skin	*Borrelia burgdorferi* (Europe)	Antibiotic trial in Borrelia-positive cases
Small bowel (IPSID)	*Campylobacter jejuni*	**α-heavy-chain disease**; early disease regresses with antibiotics
Spleen	**Hepatitis C**	**Antiviral therapy first** if no urgent indication: 7/9 CR after interferon with HCV clearance, none in HCV– (Hermine, NEJM 2002); DAAs — SVR 98%, lymphoma response **73% in MZL** (Arcaini, Blood 2016)
Parotid · thyroid	**Sjögren · Hashimoto**	Autoimmune drive — no antimicrobial; treat by stage

STAGING GASTRIC MALT — LUGANO (GI) SYSTEM

I	Confined to the GI tract (single or multiple non-contiguous lesions)
II1 / II2	Into the abdomen: **local** (perigastric) nodes / **distant** (mesenteric, para-aortic) nodes
IIE	Penetrates serosa into adjacent organs or tissues
IV	Disseminated extranodal, or supradiaphragmatic nodes. *No stage III*

Non-gastric and nodal/splenic disease: Ann Arbor/Lugano.

SMALL B-CELL DIFFERENTIAL

	CD5	CD10	CD23	Clue
MZL	–	–	–	Diagnosis of exclusion; **cyclin D1–**, MYD88 usually wild-type
CLL/SLL	**+**	–	**+**	CD200+, dim sIg / CD20; LEF1+
Mantle cell	**+**	–	–	**Cyclin D1, SOX11**, t(11;14)
Follicular	–	**+**	–/+	BCL2+ follicles, **t(14;18) IGH::BCL2**
LPL / Waldenström	–	–	–	**MYD88 L265P** (>90%), IgM, plasmacytic
Hairy cell	–	–	–	**CD11c, CD25, CD103, CD123, annexin A1; BRAF V600E**

Splenic MZL vs hairy cell: SMZL villi are short and **polar**, CD25–/annexin–/BRAF wild-type, and the marrow is intrasinusoidal and nodular; HCL has circumferential projections, a "dry tap", monocytopenia. **Splenic B-cell lymphoma/leukemia with prominent nucleoli** (WHO-HAEM5; absorbs HCL-variant) is CD25–, annexin–, TRAP–, CD123–. Splenic diffuse red pulp lymphoma is the other mimic.

MALT-IPI — 1 POINT EACH

Risk	Points	5-yr EFS
Low	0	70%
Intermediate	1	56%
High	≥2	**29%**

Age ≥70 · stage III–IV · LDH >ULN. Derived in IELSG-19 (n=401), validated in 633; prognostic in gastric and non-gastric disease (Thieblemont, Blood 2017).

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TREATMENT PATHWAY

Gastric MALT, localized

↓

H. pylori positive
eradicate: bismuth quadruple ×14 days
confirm cure (breath/stool) ≥4 wk after, PPI off

↓

Re-endoscope at ~3 months, then q3–6 mo
regression is **slow — allow ≥12 months**
persistent but stable, HP-negative: keep watching
HP still present: second-line eradication

↓

HP-neg, t(11;18)+, or failure
ISRT 24 Gy to stomach — **curative intent**
rituximab if RT not suitable

Non-gastric MALT, localized

↓

Antimicrobial where a driver fits
doxycycline (ocular), antibiotics (skin, IPSID)

↓

Then or instead
ISRT 24 Gy — curative intent
observation acceptable if asymptomatic

Splenic MZL

↓

HCV positive
DAA therapy first unless urgent

↓

Indication to treat
symptomatic spleen, cytopenias, AIHA/ITP
rituximab 375 weekly ×4 ± maintenance
splenectomy if rituximab fails or unsuitable

Nodal, disseminated, or relapsed

↓

Asymptomatic, low burden
observe (FL-like GELF logic)

↓

Needs systemic therapy
BR or rituximab ± chlorambucil
R-CHOP rarely (bulky, or concern for large cells)

↓

Relapse
zanubrutinib · **R²** · alternative chemo-immunotherapy
3L+: **liso-cel** CAR-T

SYSTEMIC THERAPY — EVIDENCE AND STATUS

Agent · US status	Evidence
Rituximab ± chlorambucil	**IELSG-19** (MALT, n=454): 5-yr EFS **68%** R-chlorambucil vs 50% rituximab vs 51% chlorambucil; PFS better; **OS ~90% in all arms**
Bendamustine-rituximab	Phase 2 MALT data (Salar, Blood 2017) and StiL NHL1 (which included MZL) support BR as the usual frontline chemo-immunotherapy for advanced disease
Obinutuzumab *not approved for MZL*	GALLIUM MZL subgroup (n=195): **no PFS benefit** (4-yr 72.6% vs 64.1%) and **more grade 5 AEs (14.9% vs 9.7%)** — use rituximab
Zanubrutinib *accelerated, 2L+ after anti-CD20 (Sep 2021)*	**MAGNOLIA** (single arm, n=68): ORR **68%**, CR 26%; 24-mo PFS 71%. AF 3%. Confirmatory **MAHOGANY** (zanubrutinib + rituximab vs R²) still ongoing; approval remains accelerated
Ibrutinib *MZL indication withdrawn Dec 2023*	PCYC-1121 ORR 48% led to 2017 accelerated approval; **SELENE (FL/MZL, + chemo) missed PFS** → withdrawn
R² *regular approval FL + MZL (2019)*	**AUGMENT** included 63 MZL; overall PFS HR 0.46 — but **the MZL subgroup showed no significant PFS benefit** (small, imbalanced). Approved nonetheless
Liso-cel *approved 3L+ (Dec 2025) — first CAR-T for MZL*	**TRANSCEND FL MZL cohort** (single arm; nodal 48%, splenic 27%, MALT 25%): ORR **95%** of 66 infused (84% of all leukapheresed, per label), CR 56% ITT; grade 3 CRS / neuro 4% each
Axi-cel *not approved for MZL*	ZUMA-5 MZL cohort: ORR 77%, median PFS not reached at 3 yr
Umbralisib, other PI3Ki *withdrawn*	Umbralisib MZL/FL approval withdrawn 2022 (excess deaths). **No bispecific is approved for MZL**

> **Transformation** to DLBCL is uncommon but prognostically important: suspect with rising LDH, a rapidly growing or FDG-hot lesion, B symptoms. **Biopsy, then treat as DLBCL**. Scattered large cells in a MALT lymphoma are not by themselves transformation — sheets are.

KEY EVIDENCE

Zullo meta-analysis	H. pylori eradication: 77.5% remission; t(11;18)+ 22%
Ferreri phase 2	Doxycycline, ocular adnexal MZL: ORR 65%
Hermine / Arcaini	HCV clearance regresses splenic MZL (IFN; DAAs 73%)
IELSG-19	R-chlorambucil > either alone for EFS; OS equal
GALLIUM (MZL)	Obinutuzumab: no PFS gain, more fatal AEs
MAGNOLIA	Zanubrutinib R/R: ORR 68%, single arm
AUGMENT	R²: benefit driven by FL, not MZL
TRANSCEND FL (MZL)	Liso-cel 3L+: ORR 95% infused, CR ~56%

AFTER ERADICATION — RESTAGING ENDOSCOPY

H. pylori	Lymphoma	Action
Gone	Gone	**Observe**: EGD q3–6 mo, then yearly
Gone	Persists	Asymptomatic and stable: **keep observing** (≥12 mo). Symptomatic or progressing: **ISRT**
Present	Gone	**Second-line eradication**
Present	Persists	Second-line eradication; **ISRT** if progressing or symptomatic

SPLENIC MZL

- **Many never need treatment** — observe asymptomatic, stable disease
- **Rituximab alone has largely replaced splenectomy** as first therapy: high response rates, no surgical risk. Splenectomy remains diagnostic and therapeutic when tissue is needed or rituximab fails
- **Before splenectomy:** pneumococcal, meningococcal (ACWY + B) and Hib vaccines ≥2 weeks before — **and before rituximab**, which blunts vaccine response
- Treat **autoimmune cytopenias** as part of the disease — they often respond to rituximab

DOSING

Regimen	Dosing
H. pylori bismuth quadruple	PPI BID + bismuth subsalicylate 524 mg QID + tetracycline 500 mg QID + metronidazole 500 mg TID–QID, **14 days**
Rituximab	**375 mg/m² weekly ×4**; maintenance 375 q8wk ×2 yr optional
BR	Bendamustine 90 mg/m² d1+2 + rituximab 375 d1, q28d ×6
R-chlorambucil *IELSG-19*	Chlorambucil 6 mg/m²/d PO wk 1–6, then 2-wk blocks (wk 9–10, 13–14, 17–18, 21–22) + rituximab 375 wk 1–4, 9, 13, 17, 21
Zanubrutinib	**160 mg BID** or 320 mg daily, continuous
R²	Lenalidomide 20 mg d1–21 q28d ×12 + rituximab weekly C1, d1 C2–5
ISRT	**24 Gy** / **12 fx**; 2 × 2 Gy for palliation

PEARLS

- **Always test for the driver** — H. pylori, HCV — before any cytotoxic therapy
- **t(11;18) predicts H. pylori non-response** (and dissemination): go to RT rather than repeated antibiotics
- **Give gastric MALT time**: histologic regression after eradication can take a year or more
- **MYD88 L265P favors LPL**; CD5+ or cyclin D1+ is not MZL